captionlabel 0em caption

aRMSD

aRMSD allows the manual the comparison of two constitutionally identical small molecule structures by the Kabsch algorithm[1]. In addition to the original approach to account only for atomic positions, aRMSD provides adjustable weights, for instance to equally account for X-ray scattering factors (presuming Mo $K\alpha$ radition). The results of the superposition can be accessed as interactive renderings by `vtk`[2] while statistics are visualized in diagrams generated by `matplotlib`.[3] The development of the software was started prior to 2016 by Arne Wagner during his PhD thesis in the Himmel group (University of Heidelberg, Germany)[4],[5] and distributed on `https://github.com/armsd/aRMSD` under the MIT license scheme.

By reorganization of the project, the principal objective of present version 1.0.0 is the provision of an easier installation and subsequent use with Python 3.10 (and above), agnostic to the operating system at hand. This is the reason why the primary input format of data to process should be `.xyz`, or (second best choice:) `.pdb` in the syntax written by OpenBabel.[6] Contrasting to prior versions of the program, correct processing of `.cif` or output files by quantum mechanical programs (Gaussian, MOPAC, etc) can not be guaranteed.

1 What may aRMSD do for you – an appetizer

As shown in the figure below, aRMSD may be used to align the models (a) and reorder the atom order by the Hungarian algorithm (b). Subsequently, the superposition is optimized by minimization of the RMSD according to the Kabsch algorithm.

[1] Kabsch, W. A Solution for the Best Rotation to Relate Two Sets of Vectors. *Acta Cryst A* **1976**, *32* (5), 922–923. `https://doi.org/10.1107/S0567739476001873`.

[2] `http://www.vtk.org`

[3] `https://matplotlib.org/`

[4] Wagner, A. Himmel, H.-J. aRMSD: A Comprehensive Tool for Structural Analysis. *J. Chem. Inf. Model.*, **2017**, *57*, 428–438. `https://doi.org/10.1021/acs.jcim.6b00516`.

[5] Wagner, A. Synthese und Koordinationschemie guanidinatstabilisierter Diboranverbindungen. (Synthesis and Coordination Chemistry of Guanidinate-Stabilised Diboranes) PhD thesis (2015), University of Heidelberg (Germany). Written in German including an English summary. The pdf of this document may be found at the doi 10.11588/heidok.00019018.

[6] Open Babel, `http://openbabel.org/wiki/Main_Page`. For further details, see by O'Boyle, N. M.; Banck, M.; James, C. A.; Morley, C.; Vandermeersch, T.; Hutchison, G. R. Open Babel: An open chemical toolbox. *J. Cheminf.* **2011**, *3*:33. `https://doi.org/10.1186/1758-2946-3-33`.

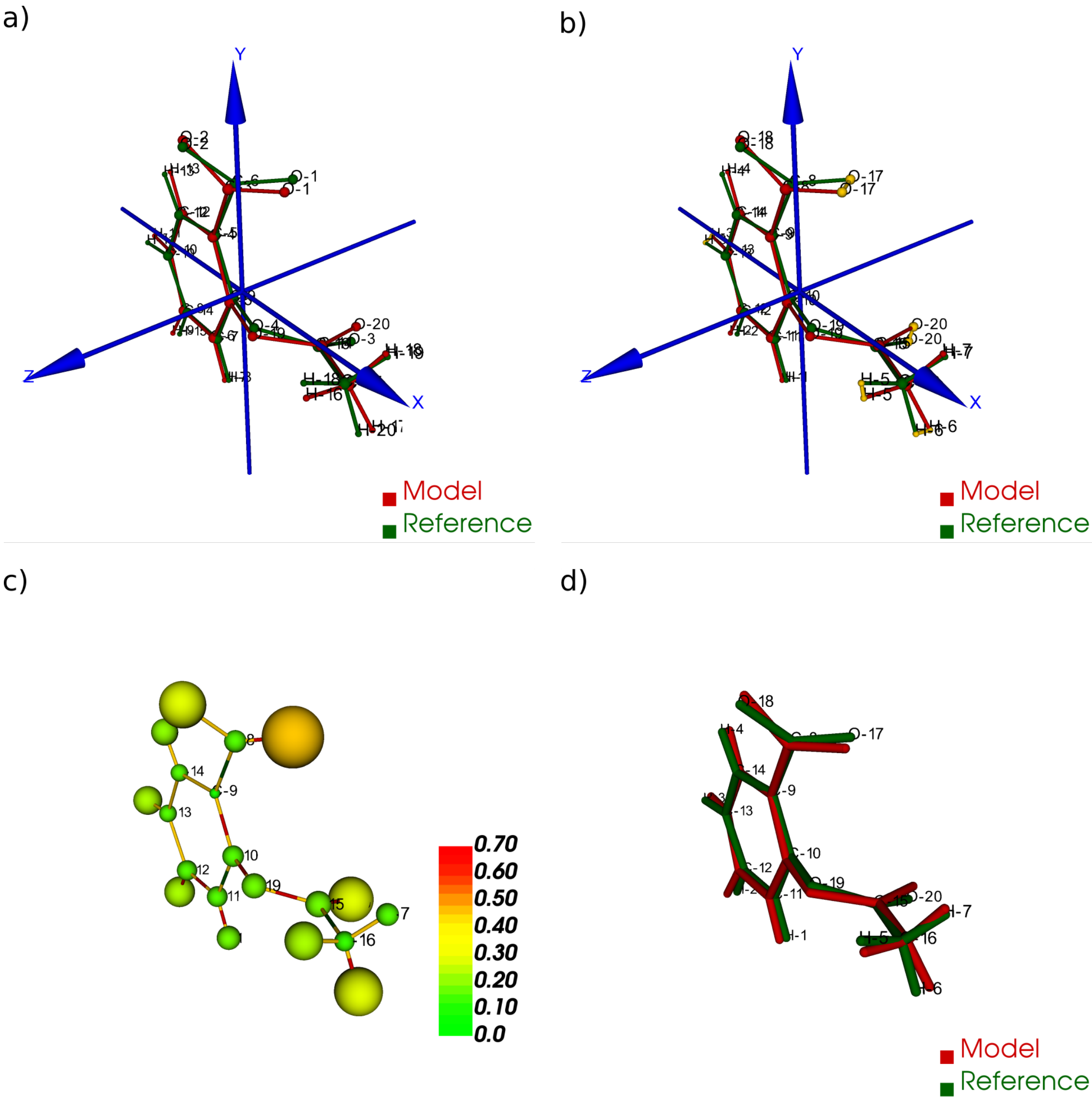

Figure 1: Subsequent stages comparing two models of the aspirinate anion with aRMSD: a) user-assisted alignment, b) re-ordering of atoms by the Hungarian algorithm, c) interactive difference rendering, d) interactive refined superposition of the models.

In a newly designed interactive representation (c), the differences between the two models are shown: atoms drawn with larger diameter indicate a larger *relative* contribution to the final RMSD determined for the complete model. Their color corresponds to the scale at the side about *absolute* positional differences (in Angstroms) of said atom in in reference and model in the optimized superposition.

In addition, `aRMSD` the program compares the corresponding bond lengths of model and reference indicates either by green or red color encoding if the one by the model is shorter, or lengthier than the corresponding bond in the reference. Not shown here, but `aRMSD` allows the interactive readout of the differences found for selected bond lengths, angles, and dihedral angles, too.

`aRMSD` equally provides a more classical (yet interactive) visualization of the optimized superposition (d). This "best fit" determined may be saved as a `.xyz` file.

In addition to a permanent log file (`aRSMD_logfile.out`) about setup and results of the superposition (e.g., final rmsd, cosine similarity, and GARD score[7]), the user may complement the scrutiny with diagrams. Generated by `matplotlib`, it is possible to pan and zoom regions of interest, and export the results for instance as `.png`, `.pdf`, or `.tikz`.

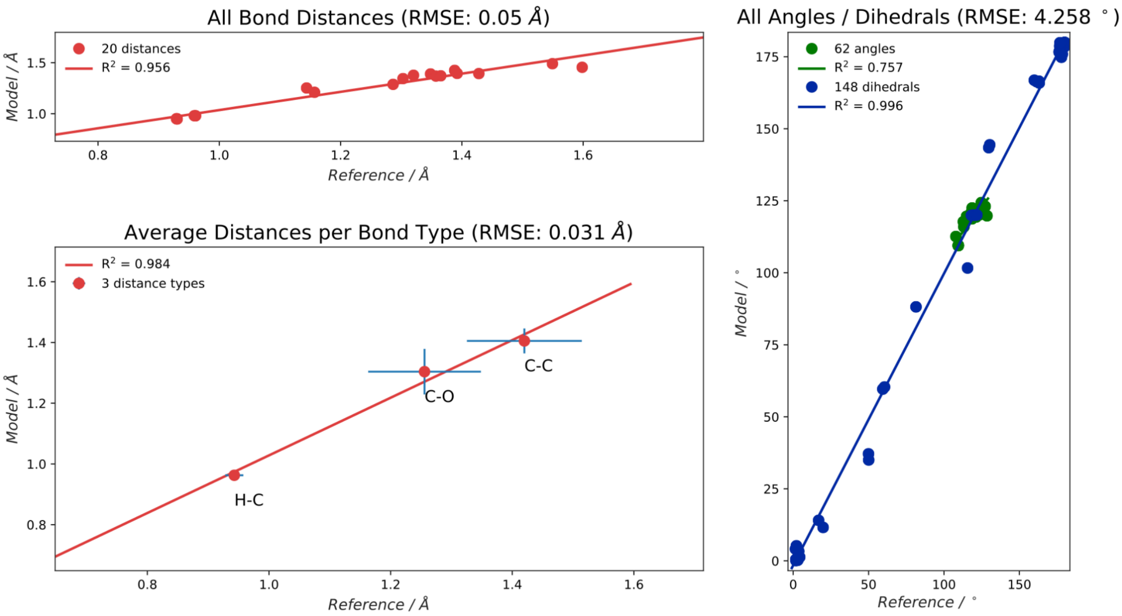

Figure 2: Statistical plots drawn by `aRMSD` about said comparison of two aspirinate model data.

[7]Baber, J. C.; Thompson, D. C.; Cross, J. B.; Humblet, C. GARD: A Generally Applicable Replacement for RMSD. *J. Chem. Inf. Model.* **2009**, *49* (8), 1889–1900. `https://doi.org/10.1021/ci9001074`.

2 Where stands aRMSD, relatively to other programs?

`aRMSD` allows a pair-wise comparison of small molecule structures with significant user-interaction. This offers you multiple levels to check and adjust the progress of the analysis. A more automatic analysis, potentially over batches of structures, is not foreseen; if interested, see for instance Jimmy Kromann's rmsd.

Note, however, that an automated unsupervised scrutiny of model data may yield wrong results. One potential pitfall is how the model information is handled prior to the refinement of the structure alignment, where `aRMSD` uses the Hungarian algorithm. To quote Kildgaard:[8]

> "The RMSD can be minimized by translating and rotating one set of coordinates (the other is held fixed) because the molecules are invariant under these operations. This will lead to the two molecules being superimposed but can also lead to a false RMSD value if the atoms are not ordered identically."

which was further demonstrated (and illustrated) for instance by Temeslo.[9]

3 Proposed installation and use

`aRMSD` preferrably is used in a virtual environment of Python 3.10 (or above) to resolve its dependencies. Depending on Python version and operating system, this support can add up to about 1GB permanent memory. With the `.whl` at disposition (release page of the GitHub repository, PyPI, or built by 'uv build' from the GitHub repository), the installation with `pip` provides `armsd` (all lower case) as new initial command to the CLI.

When running `aRMSD` from the CLI, ensure the terminal is tall enough (e.g., 40 rows instead of only 24; a width of 80 characters however is fine). Else you may miss some of its rolling interface, and commands at disposition. The `docs` folder provides you a primer.

◇

[8] Kildgaard, J. V.; Mikkelsen, K. V.; Bilde, M.; Elm, J. Hydration of Atmospheric Molecular Clusters: A New Method for Systematic Configurational Sampling. *J. Phys. Chem. A* **2018**, *122*, 5026–5036. `https://doi.org/10.1021/acs.jpca.8b02758`.

[9] Temeslo, B.; Mabey, J. M.; Kubota, T.; Appiah-Padi, N.; Shields, G. C. ArbAlign: A Tool for Optimal Alignment of Arbitrarily Ordered Isomers Using the Kuhn-Munkres Algorithm. *J. Chem. Inf. Model.* **2017**, *57*, 1045–1054. `https://doi.org/10.1021/acs.jcim.6b00546`.